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Engineering Mathematics

10Week [06 November - 12 November]

Quiz 10

Started on	2023-11-13 08:08
State	Finished
Completed on	2023-11-13 08:59
Time taken	51 mins 21 secs
Grade	5.00 out of 5.00 (100%)

Question 1

Correct

Mark 1.00 out of 1.00

Find the convolution $1 * \sin t$

Select one:

- ☐ $\cos t$
- ☐ $\cos t - 1$
- ☒ $1 - \cos t$ ✓
- ☐ $\sin t$

Your answer is correct.

The correct answer is: $1 - \cos t$

Question 2

Correct

Mark 1.00 out of 1.00

Find the Laplace transform of $\int_0^t e^v \sin(t-v) dv$

Select one:

- ☒ $\frac{1}{(s-1)(s^2+1)}$ ✓
- ☐ $\frac{s}{(s-1)(s^2+1)}$
- ☐ $\frac{s^2-s}{(s-1)(s^2+1)}$
- ☐ $\frac{s-1}{s^2+1}$

Your answer is correct.

The correct answer is: $\frac{1}{(s-1)(s^2+1)}$

Question 3

Correct

Mark 1.00 out of 1.00

Find a formula for the solution to the initial value problem

$$y'' - 9y = g(t), \quad y(0) = 2, \quad y'(0) = 0.$$

Select one:

- ☐ $\frac{1}{6}(e^{3t} - e^{-3t}) + \int_0^t (e^{3(t-v)} + e^{-3(t-v)})g(v)dv$
- ☐ $e^{3t} - e^{-3t} + \frac{1}{6} \int_0^t (e^{3(t-v)} + e^{-3(t-v)})g(v)dv$
- ☒ $e^{3t} + e^{-3t} + \frac{1}{6} \int_0^t (e^{3(t-v)} - e^{-3(t-v)})g(v)dv$ ✓
- ☐ $\frac{1}{6}(e^{3t} + e^{-3t}) + \int_0^t (e^{3(t-v)} - e^{-3(t-v)})g(v)dv$

Your answer is correct.

The correct answer is:

$$e^{3t} + e^{-3t} + \frac{1}{6} \int_0^t (e^{3(t-v)} - e^{-3(t-v)})g(v)dv$$

Question 4

Correct

Mark 1.00 out of 1.00

Evaluate the integral $\int_{-1}^1 (\cos 2t)\delta(t)dt.$

Select one:

- ☐ -1
- ☐ 0
- ☒ 1 ✓
- ☐ ∞

Your answer is correct.

The correct answer is: 1

Question 5
Correct
Mark 1.00 out
of 1.00

Determine the Laplace transform of $t^3\delta(t-3)$.

Select one:

- ☒ $27e^{-3s}$ ✓
- ☐ $27e^{3s}$
- ☐ 27
- ☐ e^{3s}

Your answer is correct.

The correct answer is: $27e^{-3s}$